THE UNIVERSITY OF BRITISH COLUMBIA

SAMPLE FINAL EXAMINATION

Common design principles: Visibility, Feedback, Constraints, Mapping, Consistency, Affordance, Simplicity, Matching, Minimizing memory load, Diagnose/Recover errors, Control and freedom, Flexibility, Provide Help

Names mentioned in the history lecture: Vannevar Bush, Douglas Engelbart, Ivan Sutherland, Alan Kay

Usability Goals: Effectiveness, Efficiency, Safety, Utility, Learnability, Memorability.

PART I – True and False (There will be ~8-10 questions in the final exam)

Circle either True or False for each of the following statements::

1) True False 20/20 vision means that what a person with normal vision sees at 20 feet, a test subject sees at 20 feet

2) True False Mouse is an example of an absolute input device

3) True False Fitt's law is used to predict users' typing performance.

4) True False Closure principle states that elements arranged in a straight line or a smooth curve are perceived as being more related

5) True False Systems that require the least time or less number of steps are usually considered to be less efficient

7) True False Color is selective and associative

8) True False A letter with a serif font commonly includes a small shape or projection at the beginning or end of a stroke.

9) True False Android Bundle is used to pass data between activities

Part II – Multiple Choice (There will be ~8-10 questions in the final exam)

Circle only **one letter** for each question:

1. ______________ light sensors in our eye help us to perceive colours.
   a. Rods
   b. Cons
   c. Both
   d. None of the above

2. When selecting a button with a mouse, the larger the distance, the _____________ the time.
   a. Shorter
   b. Longer
   c. Equal
   d. None of the above

3. Multi-touch surfaces on smartphones provide ___________ manipulation where a single finger contact on the surfaces with __________
    a. Direct manipulation, 1 DOF
    b. Indirect manipulation, 1 DOF
    c. Direct manipulation, 2 DOF
    d. Indirect manipulation, 2 DOF
    e. None of the above

4. _________________ principle describe that individual elements are associated more strongly with nearby elements than with those further away
    a. Similarity
    b. Continuity
    c. Proximity
    d. Closure
    e. None of the above

5. The colour model most commonly used in printed materials is:
    a. HSV/HSB
    b. RGB
    c. CMY/CMYK
    d. All of the above
    e. None of the above

6. Prototyping that evaluates one aspect of the system in depth is called a:
    a. High-fidelity prototyping
    b. Horizontal prototyping
    c. Vertical prototyping
    d. Low-fidelity prototyping
    e. None of the above

7. There is exactly one target inside each region, and that target is the closest target to any point within that region. In mathematical terms, this is referred to as a

    ____
    a. Area cursor
    b. Basic cursor
    c. Bubble Cursor
    d. Voronoi Diagram
    e. None of the above

8. To support the up functionality (i.e., back to the app's main screen) in an activity, you need to declare the activity's parent. You can do this in the ________________, by

setting ________________ attribute.

a. App manifest, android:parentActivityName
b. App Java file, MainActivity.Java
c. App Layout, activity_main.xml
d. App values, Strings.xml
e. None of the above

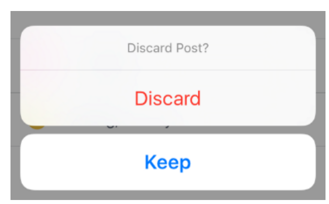
Part III - Short Answer (There will be ~15-20 questions in the final exam)

1. Occlusion is a common problem of touchscreen interactions, especially on small-screen devices. Explain one technique that could be used to reduce occlusion while showing occluded items and allowing users to access them even if they are located at the edge of a screen.

   Technique name: Shift (however, if students forget the name of the technqieu and write something similar to shift such as "make occlusion area visible", "pop-up", "pop-out" they will get marks)

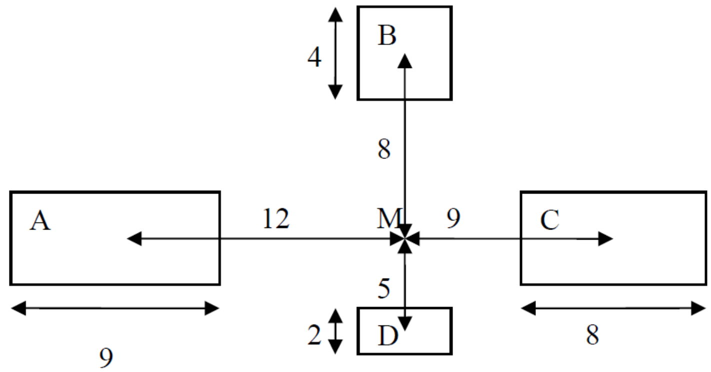
   How it resolves occlusion issue: The technique shows a copy of the occluded screen area and places it in a non-occluded location

2. A social media application shows the following dialogue box if a user taps on "Cancel" by mistake. The design principle from the list below ________________can be best illustrated with this example.



   Visibility, Constraints, Mapping, Consistency, Simplicity, Matching, Minimizing memory load, Control and freedom, Flexibility

3. Let's assume that you have four targets A, B, C and D displayed on a desktop computer screen (see the image below) where a mouse cursor is located at M. Assume a user can move his/her hand equally well in all target directions. None of the targets are located close to the edge of the screen. Starting at location M, which target is the fastest to reach with the mouse? Show your work.



Your answer:

MT = a + b $\log_2(1 + D/W)$. For mouse, a and b will be the same in all conditions. Therefore, we can only use D/W

A → (D/W = 1 1/3)

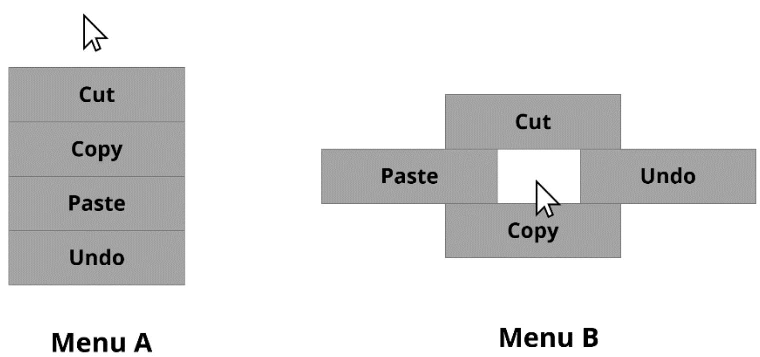
B → (D/W = 2)

C → (D/W = 1 1/8)

D → (D/W = 2 1/2)

C is faster to reach

4. You need to decide between two menu designs, Menu A and Menu B, as shown below. Assume that the menus are opened once you press the right mouse button. The initial position of the cursor is shown below for both menus. Assume that users are required to select operations (i.e., cut, copy, paste, undo) equally in a test session.



Which menu will be the fastest to use (on average)? Explain why (you don't need to show any calculation).

Fastest Menu: Menu B
Explanation:

All items are close (same distance) to the cursor in Menu B. Some are far away in Menu A. Fitts law says that items that are farther away take more time to select.

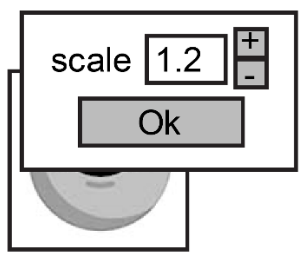
5. Use examples to explain the difference between a typeface, and a font?

   Typeface refers to a family of fonts sharing the same name, like “Times New Roman.”
   A font is a choice of typeface and size and style, like “Time New Roman, 12pt, italic”

6. What is perceived affordance? Give an example of a good affordance in a user interface.

   A visual property of an object that suggests how it should be used. One example could be a resized grip handle on a window; another could be blue underlined text that has become an affordance for a clickable link.

7. The interface below is designed to adjust the scale of an image in a sketch program. The scale can be adjusted in a dialog using (a) the numeric textbox or (b) using “up/down” buttons (Note that the scale is uniform). Pressing the “Ok” button applies the scale to the image.



   Use the Keystroke Level Model (KLM) to evaluate the designs. Assume that (1) the hand is on the mouse, (2) the dialog is already open, (3) the current scale value is 1.2, and (d) the cursor is located outside the dialog box. Here are the KLM operators:
   K = 0.3s
   P = 1.1s
   B = 0.1s
   H = 0.4s
   Note: only use B for a click (instead of BB) and you do not include M in your sequence and time calculation.

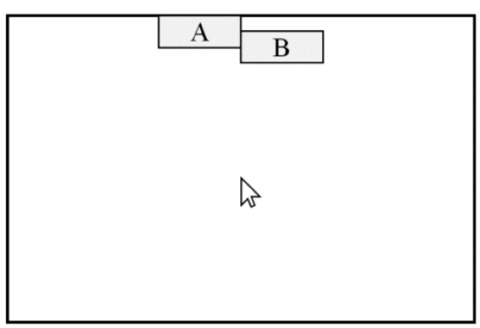
Provide the sequence of KLM operators and calculate the total time to change the scale to 0.8. Assume that the entire text (i.e., 1.2) is selected when a user clicks on the textbox. The user types the desired value (i.e., 0.8) with the keyboard and then uses the mouse to apply the new scale to the picture by clicking the Ok button.

The sequence of KLM operators:
Move the cursor on top of the textbox – P
Press the mouse button to highlight "1.2" - B
Move the hand from the mouse to the keyboard – H
Enter "0.8" - KKK
Move the hand from the keyboard to the mouse - H
Move the cursor on top of the ok button – P
Press the ok button - B

8. Using a mouse to point (a = -11 and b = 6), what is the movement time to click on an 120-pixel by 100-pixel Cancel button located 300 pixels away?

$MT = -11 + 6 * \log_2(300/100+ 1)$
$= -11 + 6*\log_2(4)$
$= -11 + 6 * 2$
$= -11 + 12$
$= 1$ sec

9. Assume that a mouse cursor is located in the middle of a screen (see the picture below). On average, which button will be the fastest to select, A or B? Explain why, and be sure to reference Fitts' Law in your explanation (you don't need to calculate anything).



Which button will be faster to select? A will be faster to select

Explanation: A has infinite target width (as it's located at the edge, a user can just shoot the mouse to that direction. Thus, A will be easier to select. If they even apply fitts law, D/W, where for target A, width is infinity. So, A is easier to select.

If they say B is faster, check their explanation. I would give them 1 mark if they could provide good justification for B

10. DJ Mike has just created a new rhythm-based game. He would like to run an experimental study to examine two different game controllers and see which will provide a more compelling game play experience. One controller (on the left, below) is a mock turntable, and the other controller (on the right, below) is a standard Wii Mote.

To compare the two interfaces, Mike has designed the following experimental study:

- First, all participants will play five games using the turntable interface.
- They will then have a three-minute break.
- After the break, all participants will play the same five games using the Wii Mote.
- Error rates will be measured for every game to determine the best interface.
- Subjects will also be asked to rate on a scale of 1 to 5, which interface they prefer.

If subjects experienced only one interface or condition, this would be known as a ____________________ subjects design study.

Your answer: between

Write one independent variable used in this study. What are the levels?

Independent variable: Interface Type or Game Controller

Levels: Wii Mote or turntable interface

What two dependent variables used in this study?

Dependent variable 1: Error Rate

Dependent variable 2: User Rating

11. William was interested in the effects of caffeine on sleep. He randomly assigned university students (10 men and 10 women), all healthy and of average weight (e.g., 75lb), to four groups. One group (i.e., Group A) drank decaffeinated coffee, and the other three groups, Group B, C, D, drank one, two, or three cups of coffee, respectively. He found that people took longer to fall asleep after they drank more coffee. Based on this study, provide one example of each of the following variables:

Dependent variable: Time taken to fall asleep

Independent variable: Amount of coffee consumed by the participants

Control variable: All university students; average weight; healthy, etc.